

FedBook-Sem

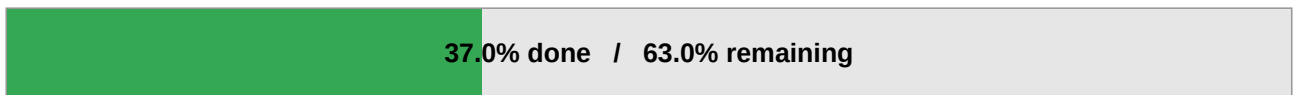
Development Status Report

Federated Social Bookstore + ML Book Recommender

Scope: development work only - excludes evaluation and thesis writing

Project:	R26-IT-107 / IT22922670 (SLIIT B.Sc. FYP)
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Effort unit:	person-hours (midpoint estimates)

Overall progress



Total scope estimated: 994 h

Delivered so far: 368 h (37.0%)

Remaining to build: 626 h (63.0%)

Estimates are midpoints of realistic development ranges (not best-case). They include debugging, integration pain, and reasonable iteration cycles, but exclude: supervisor meetings beyond the ~20h allocated, dataset licence surprises, and long waits for external approvals (ethics, API keys).

Effort summary by component

Component	Done (h)	Remaining (h)	% done
Existing federated bookstore (delivered)	368	0	100%
Phase 1 finish	0	24	0%
Phase 2 - Semantic engine	0	80	0%
Phase 3 - ML Models	0	110	0%
Phase 4 - Cross-platform ingestion + sentiment	0	160	0%
Phase 5 - Novel contribution (sentiment-aware + learned fusion)	0	100	0%
Phase 6 - Neo4j GDS variant (optional but distinctive)	0	42	0%
Phase 8 (dashboard only, thesis writing excluded)	0	40	0%
Cross-cutting development work	0	70	0%
TOTAL	368	626	37%

What has been delivered

The federated social bookstore platform is complete and running. In addition, the Python ml-service package skeleton (config, Neo4j client, Open Library client, entity resolver, tests, fixture, Dockerfile) has been scaffolded in the current session but is not yet verified end-to-end in a container.

Delivered work	Hours
Backend: Express app, 10 routes, JWT + bcrypt	80
Neo4j driver + schema + graph query helpers	40
ActivityPub federation (actors, WebFinger, inbox, HTTP sigs, delivery)	60
W3C Web Annotations (selectors, threading, per-book queries)	20
Book-cover SVG generator + avatar uploads (multer)	10
React frontend: 6 pages, book grid, feed, annotations, JWT client	90
Docker Compose infra + seed data + ER/function diagrams + API docs	30
Integration testing / debugging / handover	20
ml-service scaffolding this session (config, neo4j client, OL client, resolver, tests, fixture)	15
docker-compose.yml: pinned Neo4j 5.15, added ml-service block	2
backend/src/graph/schema.js: openLibraryWorkId index	1
Subtotal delivered	368

Remaining: Phase 1 finish

Phase subtotal: 24 h (3.8% of remaining scope)

Task	Hours
Verify Docker build + first-run pytest inside container	4
Apply Neo4j schema (npm run schema) + verify FBS regression-free	3
Load Kaggle 7k books.csv (after user download)	3
Entity resolver >=90% accuracy tuning + fixture expansion if needed	8
Windows/Docker path + TLS + bind-mount debugging	6
Phase subtotal	24

Remaining: Phase 2 - Semantic engine

Phase subtotal: 80 h (12.8% of remaining scope)

Task	Hours
sentence-transformers module + tests	10
Neo4j vector index creation + verification	4
build_embeddings.py: embed ~6810 books, write vectors back	10
Vector search wrapper (db.index.vector.queryNodes) + tests	8
FastAPI app: /health, /recommend/similar (isbn + text seeds)	16
Container lifespan, dependency injection, uvicorn CMD	6
Node backend proxy /api/feed/recommendations/ml	8
Frontend: getMLRecommendations() + FeedPage swap + fallback logic	12
End-to-end integration testing (Python <-> Node <-> React)	6
Phase subtotal	80

Remaining: Phase 3 - ML Models

Phase subtotal: 110 h (17.6% of remaining scope)

Task	Hours
Download + unpack goodbooks-10k	2
LightFM data prep + item feature engineering	14
LightFM WARP training loop + hyperparameter tuning (multiple runs)	24
Goodbooks-book_id -> :Book ISBN join script	8
FastAPI /recommend/cf endpoint	6
BGE cross-encoder integration + batching + model download handling	12
API extension for ?reRank=cross-encoder + latency budget testing	10
Regression tests + golden fixtures	8
Optional: fine-tune all-MiniLM on genre pairs (stretch)	26
Phase subtotal	110

Remaining: Phase 4 - Cross-platform ingestion + sentiment

Phase subtotal: 160 h (25.6% of remaining scope)

Task	Hours
Reddit PRAW client + OAuth + rate limit handling + subreddit allowlist	22
YouTube Data API v3 + quota management + 30-day expiry stamping	24
Bluesky AT Protocol client + search	16
Mastodon REST client + instance allowlist	14
Base Collector protocol + entity-resolver wiring	8
CardiffNLP twitter-roberta sentiment (with Twitter preprocessing)	12
PyABSA aspect-based sentiment setup	14
:PlatformReception graph model + schema constraint	4
Daily ingest worker + cron scheduling	8
YouTube 30-day purge cron	4
Reddit deletion sweep (periodic re-poll)	6
Per-collector tests with mocked fixtures	16
Ethics approval paperwork + API key onboarding	12
Phase subtotal	160

Remaining: Phase 5 - Novel contribution (sentiment-aware + learned fusion)

Phase subtotal: 100 h (16.0% of remaining scope)

Task	Hours
reception.py: per-platform positive share, weighted score, entropy diversity	14
rerank.py: linear alpha/beta/gamma blend + validation	6
ml/ltr.py: feature builder (6-feature vector) + LightGBM LambdaRank	22
Training script for LTR (consumes held-out data)	10
API ?reRank=linear learned with per-candidate feature echo	8
Ablation logging (JSONL per request)	4
Frontend ReceptionBadges.jsx component + per-platform colour dots	10
Debug feature-breakdown toggle in dev builds	4
Regression + fixture-driven ordering tests	8
Iteration cycles as offline results come in (5-10 rounds)	14
Phase subtotal	100

Remaining: Phase 6 - Neo4j GDS variant (optional but distinctive)

Phase subtotal: 42 h (6.7% of remaining scope)

Task	Hours
Install GDS plugin in compose + security config + restart	4
graph_search.py: gds.knn + gds.pageRank wrappers	16
API /recommend/graph?strategy=knn ppr	8
Node backend proxy + frontend hook (optional)	6
Comparison tests: native vector vs gds.knn agree	8
Phase subtotal	42

Remaining: Phase 8 (dashboard only, thesis writing excluded)

Phase subtotal: 40 h (6.4% of remaining scope)

Task	Hours
Streamlit app scaffold + docker-compose entry (:8501)	8
Recommender demo page (search, results, badges, feature contributions)	16
Model registry page (loaded artefacts, sizes, metadata)	8
Screenshots for supervisor/defence demo	4
Reproducibility guide (docker-only workflow)	4
Phase subtotal	40

Remaining: Cross-cutting development work

Phase subtotal: 70 h (11.2% of remaining scope)

Task	Hours
Supervisor iteration cycles + rework	20
Docker networking / Windows path / cache debugging	14
Multi-layer integration bugs (Py <-> Node <-> React <-> Neo4j)	16
API key onboarding lag (Reddit + YouTube approvals)	8
README + docs updates as scope evolves	8
Deployment / hosting decisions (HF Spaces? local?)	4
Phase subtotal	70

Bottom line

Total development scope for this project (existing platform + ML additions, development work only) is estimated at 994 hours.

Of that, 368 hours (37.0%) has been delivered so far. 626 hours (63.0%) of development work remains.

At 40 h/week that is 15.7 weeks of full-time work still to do on development alone -- before counting evaluation code, human study, thesis chapters, or defence preparation, which are explicitly OUT OF SCOPE of this report.

Not included in these numbers

- Offline evaluation code (P@k, R@k, NDCG@k, MAP@k, novelty, diversity, coverage)
- 7-strategy sweep runner and ablation notebook
- Human face-validity study (recruit ~15-20 respondents, forms, Kendall tau)
- Thesis chapters (literature review, methods, results, discussion, ethics)
- Defence slide deck + rehearsal
- Reference formatting and appendices

If included, these would add roughly 240-300 hours (~30% more) on top of the figures in this report.

Development-only risks that can push cost higher

- Reddit / YouTube API-key approval delays (days-to-weeks of calendar loss)
- Novel-contribution rework if the learned fusion loses to baselines in early tests (may need to pivot the Phase 5 approach)
- Windows / Docker / TLS / bind-mount debugging (already observed one TLS timeout)
- Model training tuning cycles (LightFM and LightGBM often need 5-10 iterations)
- UCSD Book Graph licence handling for offline use
- Multi-layer integration bugs across Python <-> Node <-> React <-> Neo4j

Recommended contingency: +15% on the remaining figure (~93 h) to protect against the above.